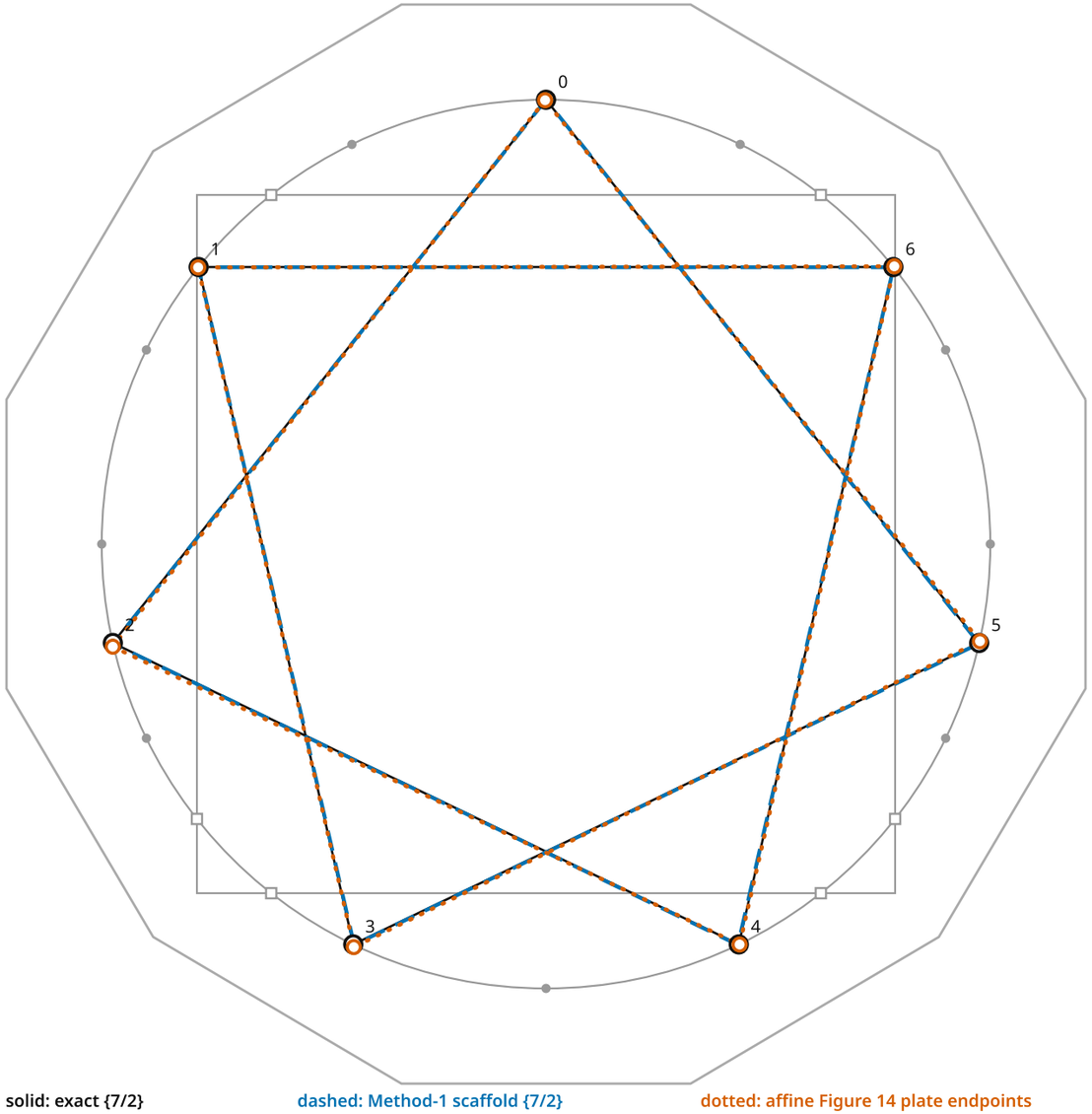
Exact-heptagon comparison and one-trisection synthesis

Fixed north orientation; no rotational fitting or endpoint reassignment

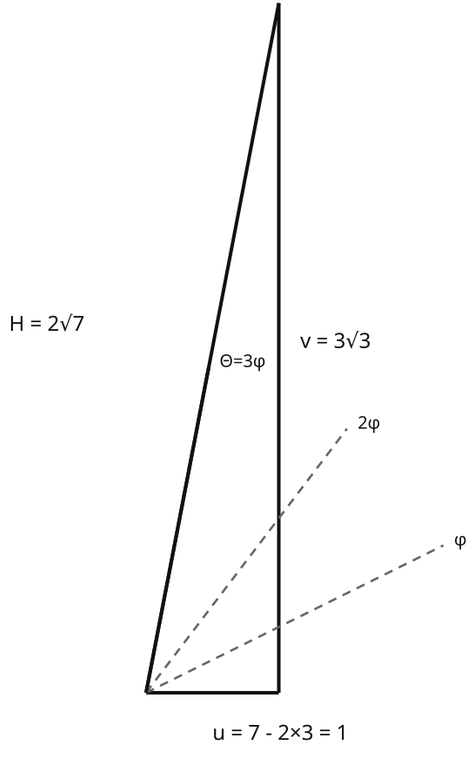
Exact sevenfold within the New Jerusalem architecture



solid: exact {7/2} **dashed: Method-1 scaffold {7/2}** **dotted: affine Figure 14 plate endpoints**

outer wall / Moon centres / square-circle junctions: context only

Native anchor and the single cubic-capable operation



Exactly one operation: TRISECT_ANGLE(Θ) → φ

$z = (2\sqrt{7}/3)\cos\varphi$; $y = z - 1/3$

$y^3 + y^2 - 2y - 1 = 0$

unique root $1<y<2 \rightarrow y = 2\cos(2\pi/7)$

Method-1 angular RMS vs exact: 0.084260°

Figure 14 plate angular RMS vs exact: 0.224059°

Source consistency only; no refit and not independent validation.